ALICE L WALTON SCHOOL OF MEDICINE

Cholinergics and Anticholinergics–2/2

Last updated October 27, 2025 10 min read ScholarRx

Adapted by: Diptiman Bose, Trager Hintze, Daedelus Hoffman

Editors: Jordan Finn, Russell Johnson

Adapted from: Muscarinic Antagonists

Listen to this Brick
0:00 / 18:45 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Predict the organ-specific effects resulting from activation and inhibition of cholinergic receptors based on the tissue expression patterns.
2. Explain the therapeutic uses of cholinergics and anticholinergics based on the [illegible].
3. Compare and discuss the adverse effects of various anticholinergic agents.

CASE CONNECTION

NK is a 76-year-old male with severe lower abdominal pain. “I haven’t peed in several days, and I feel like I might explode,” he says. He has never had trouble like this before and is generally healthy. He has been taking an over-the-counter allergy pill recently because of a runny nose. On exam, you note a markedly distended bladder; you quickly and carefully empty his bladder by catheterization. He immediately feels better and thanks you profusely. “I don’t want to go through that ever again,” NK says. “What caused this?”

What will you tell NK about the mechanisms behind urinary retention? Consider your answer as you read, and we’ll revisit NK at the end of the brick.

GO TO CONCLUSION ↓

What Are Anticholinergics? (LO#1)

Anticholinergics (or Antimuscarinics or Muscarinic antagonists) are drugs that inhibit the actions of muscarinic acetylcholine receptors found throughout the human body. For this reason, they are referred to as **anticholinergic agents** (Figure 1). Remember that acetylcholine receptors are divided into nicotinic and muscarinic receptors. The nicotinic receptors are usually found between first-order and second-order neurons in the autonomic nervous system. Muscarinic receptors [illegible] nd-order parasympathetic neurons and their target organs (e.g., smooth muscles).

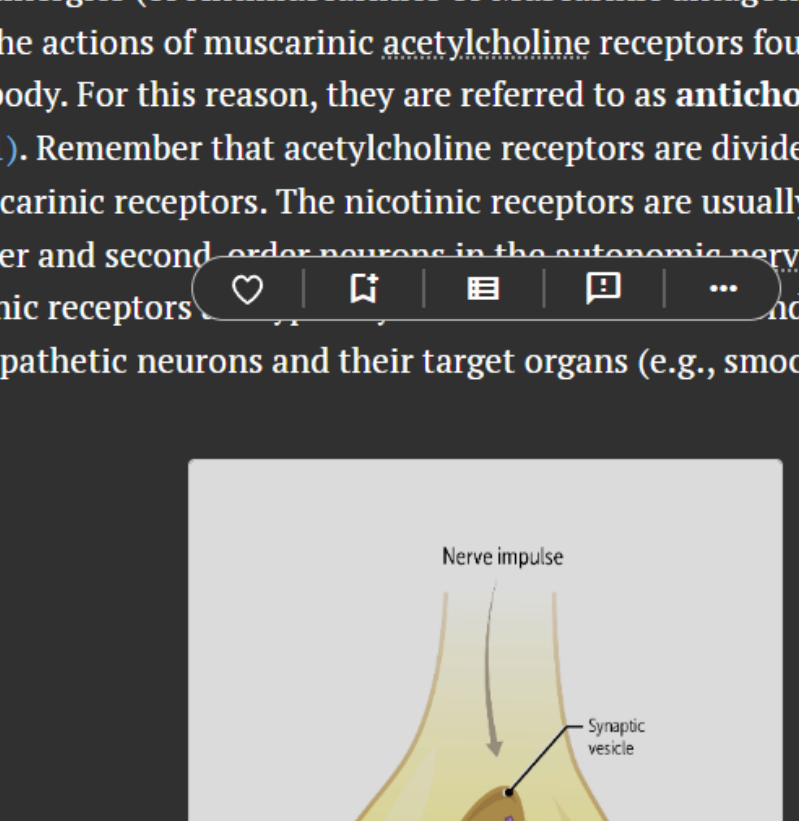


Figure 1

What Happens When Cholinergic Receptors Are Blocked? (LO#1)

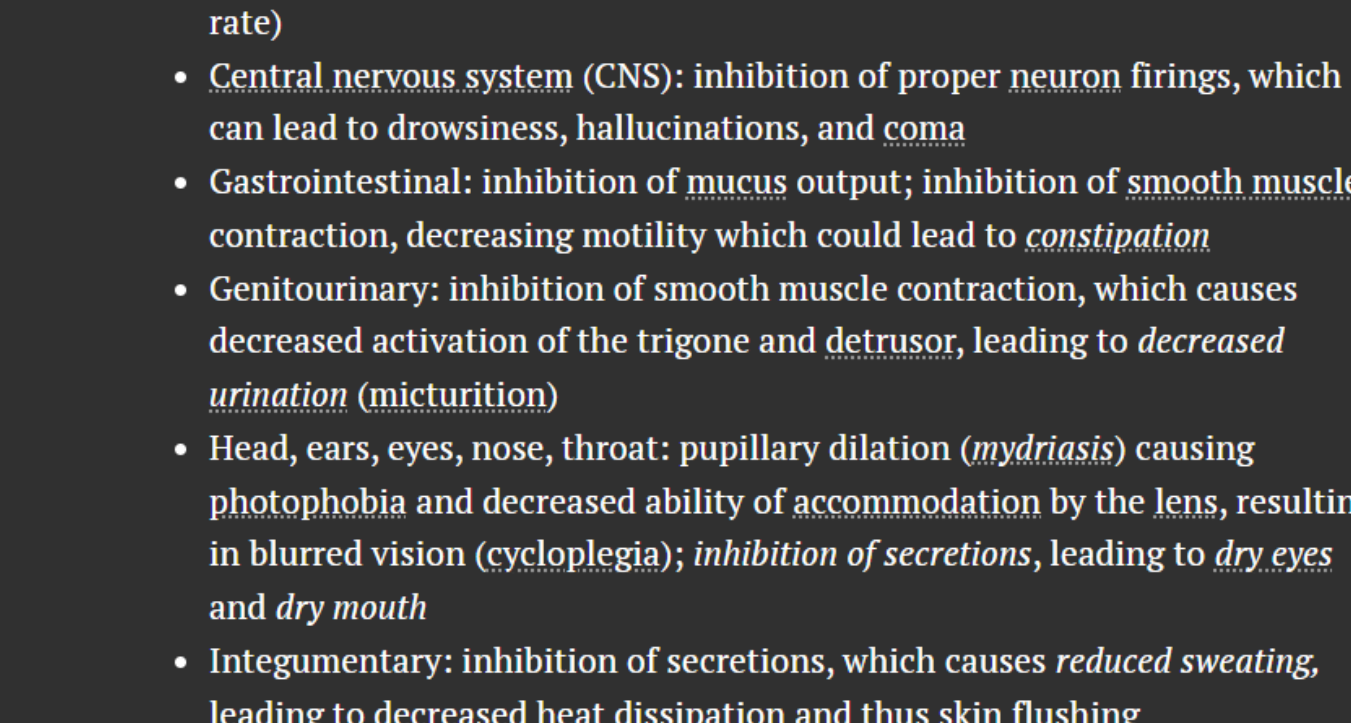
Well, to understand that, just look at how many systems the muscarinic blockade affects. Recall from the ANS-review and cholinergics and anti-cholinergics (1 of 2) [illegible] ffects are increasing glandular secretions (saliva, sweat, gastric, intestinal), increasing gastrointestinal (GI) motility, opening urinary and GI sphincters, slowing the heart rate, and constricting pupils. These functions are generally needed for eating and rejuvenating the body, as opposed to the fight-or-flight responses of the sympathetic nervous system.

What effect does PSNS activation have on the urinary bladder?

Anticholinergics act primarily through **competitive inhibition** of muscarinic receptors in the peripheral nervous system (PNS), CNS, or both. As with the cholinergics, most of the anticholinergics do not demonstrate receptor selectivity for the muscarinic subtypes and they do not block the nicotinic receptors. Figure 2 explains the effects that anticholinergics have on the various organs. Needless to say, the effects are opposite to what we learned about cholinergic effects.

Let's break down an[illegible]k at the effects:

- Cardiac: inhibition of parasympathetic inhibitory mechanisms, leading to tachycardia but useful in patients with *bradycardia* (abnormally slow heart rate)
- Central nervous system (CNS): inhibition of proper neuron firings, which can lead to drowsiness, hallucinations, and coma
- Gastrointestinal: inhibition of mucus output; inhibition of smooth muscle contraction, decreasing motility which could lead to *constipation*
- Genitourinary: inhibition of smooth muscle contraction, which causes decreased activation of the trigone and detrusor, leading to *decreased urination* (micturition)
- Head, ears, eyes, nose, throat: pupillary dilation (*mydriasis*) causing photophobia and decreased ability of accommodation by the lens, resulting in blurred vision (cycloplegia); *inhibition of secretions*, leading to *dry eyes* and *dry mouth*
- Integumentary: inhibition of secretions, which causes *reduced sweating,* leading to decreased heat dissipation and thus skin flushing
- Pulmonary: inhibition of secretions, leading to *decreased mucus production*; inhibition of smooth muscle contraction, leading to *bronchodilation* (useful in treating asthma)



ANTICHOLINERGIC EFFECTS

ORGAN	EYES	NASAL	SALIVARY GLANDS	HEART	LUNGS	GI/INTESTINE	URINARY BLADDER	SKIN
EFFECT	Mydriasis (dialate), Decreased lacrimation	Decreased secretions (dryness)	Decreased secretions (dryness)	Increased heart rate	decreased bronchial secretions	[illegible]d GI [illegible]ns, [illegible]eased vomiting, decreased intestinal motility (constipation)	Decreased urination	Decreased sweating
RECEPTOR	M3	M3	M3	M2	M3	M3	M3	M3

Figure 2

CLINICAL CORRELATION

Many of the anticholinergics used regularly in clinical practice include the letters “trop” in their name, reminiscent of the drug atropine, named after *A belladonna*, which in turn is named after Atropos, one of the Three Fates in Greek mythology. She was the Fate who cut the thread of life for mortals, a fitting name for a plant associated with severe poisoning.

ACTIVE LEARNING QUESTION

Which effect is caused by anticholinergics?

A. ● Decreased glandular secretions
B. ○ Increased GI motility
C. ○ Increa[illegible]d glandular secretions
D. ○ Constricted pupils

Explanation

The correct answer is decreased glandular secretions (A). Anticholinergics inhibit muscarinic receptors, leading to reduced secretions. Increased GI motility (B) and increased glandular secretions (C) are effects of cholinergics, not anticholinergics. Constricted pupils (D) are also a cholinergic effect.

What Are Common Anticholinergics and Their Clinical Uses?

Modern examples of anticholinergics (most with the “trop” label) include atropine and scopolamine, which are both naturally extracted from plant alkaloids. Others do not carry the “trop” label include oxybutynin, trihexyphenidyl, and tolterodine.

- **Atropine** is the oldest and most established of these agents, with many indications. It reduces respiratory secretions prior to the induction of anesthesia or surgical procedures. It improves cardiac conduction in sinus bradycardia and atrioventricular node block (Inhibition of M2 block). It is an antidote for acetylcholine toxicity (eg, poisoning by organophosphates or muscarine-containing wild mushrooms-will be discussed next week). During your ophth[illegible]ced that they use a drop to dilate your pupils prior to an examination. Atropine (and **tropicamide**) drops may be used to dilate the pupils for ophthalmoscopic exams. The dilation of the pupils helps for retinal and pupillary eye examination.
- **Scopolamine** (**hyoscine**) is usually formulated as a transdermal patch and used to *treat motion sickness and postsurgical nausea*. The patches are not for continuous use and should be replaced after 72 hours.
- **Inhaled Anticholinergics**: If you recall, presence of M3 receptors in the lungs can cause bronchoconstriction. Anticholinergics such as **ipratropium** and **tiotropium** are prescribed for patients with *asthma and chronic obstructive pulmonary disease (COPD)* to reduce mucus secretions and decrease the bronchoconstriction and airway resistance. This improves airway resistance and allow for easier respiration. These agents are also formulated in combination with β-agonists and corticosteroids to provide bronchodilation.
- **Oxybutynin** is the prototypical anticholinergic that is used in the treatment of *urinary incontinence (overactive bladder)*. Other agents that have similar effects are **tolterodine, darifenacin, solifenacin, and fesoterodine.** These agents are used in the management of urge incontinence and bladder detrusor muscle overactivity. Their slowing of bladder contractions (M3) helps patients better control their urine output.

ACTIVE LEARNING QUESTION

Which anticholinergic is used to treat motion sickness?

A. ○ Atrop[illegible]
B. ● Scopolamine
C. ○ Ipratropium
D. ○ Tiotropium
E. ○ Oxybutynin

Explanation

The correct answer is scopolamine (B). Scopolamine is typically used in the form of a transdermal patch to treat motion sickness and postsurgical nausea. Atropine (A) is used for respiratory secretions, cardiac conduction, and pupil dilation. Ipratropium (C) and tiotropium (D) are used for asthma and COPD. Oxybutynin (E) is used for urinary incontinence.

CLINICAL CORRELATION

Urge incontinence is sometimes called “overactive bladder.” Patients experience a sudden urge to urinate and then have an involuntary loss of urine for no apparent reason. The cause is unknown, but caffeine, constipation, obesity, and diabetes worsen the problem. Bladder training and pelvic floor exercises can help the patient control urine output.

What Are the Adverse Effects of Anticholinergics? (LO#3)

As with many medication classes, MAs have vastly variable CNS penetration. This is why certain MAs are associated with a greater risk of adverse CNS effects (eg, benztropine, atropine) and also why certain MAs (eg, ipratropium, glycopyrrolate) are not useful in treating CNS diseases, such as Parkinson disease.

The adverse effects of these drugs can be predicted from the list of expected receptor effects outlined previously in Figure 3. A classic mnemonic encompasses the many effects (and adverse effects) of anticholinergics: mydriasis, photophobia, and cycloplegia from their effects on eyes (blind as a bat); tachycardia and decreased heat dissipation leading to overheating (hot as a hare); the general inhibition of secretions in the mucous membranes and skin (dry as a bone); the flushed skin that results from increasing central temperature (red as a beet); the decreased excretion of wastes, both urine and stool, due to decreased motility (full as a flask); and the many CNS effects a patient might experience, including drowsiness, hallucinations, and coma (mad as a hatter). (Figure 3)

ACTIVE LEARNING QUESTION

Which anticholinergic effect is described as 'mad as a hatter'?

A. ○ Mydr[illegible]
B. ○ Tachycardia
C. ○ Decreased secretions
D. ● CNS effects

Explanation

The correct answer is CNS effects (D). 'Mad as a hatter' refers to the CNS effects such as drowsiness, hallucinations, and coma. Mydriasis (A) refers to 'blind as a bat'. Tachycardia (B) refers to 'hot as a hare'. Decreased secretions (C) refers to 'dry as a bone'.

Mnemonic

Remember adverse effects of muscarinic antagonists (blurry vision, overheating, dry mouth, flushed skin, urinary retention, and confusion) this way: **Blind** as a Bat, **Hot** as a Hare, **Dry** as a Bone, **Red** as a Beet, **Full** as a Flask, **Mad** as a Hatter

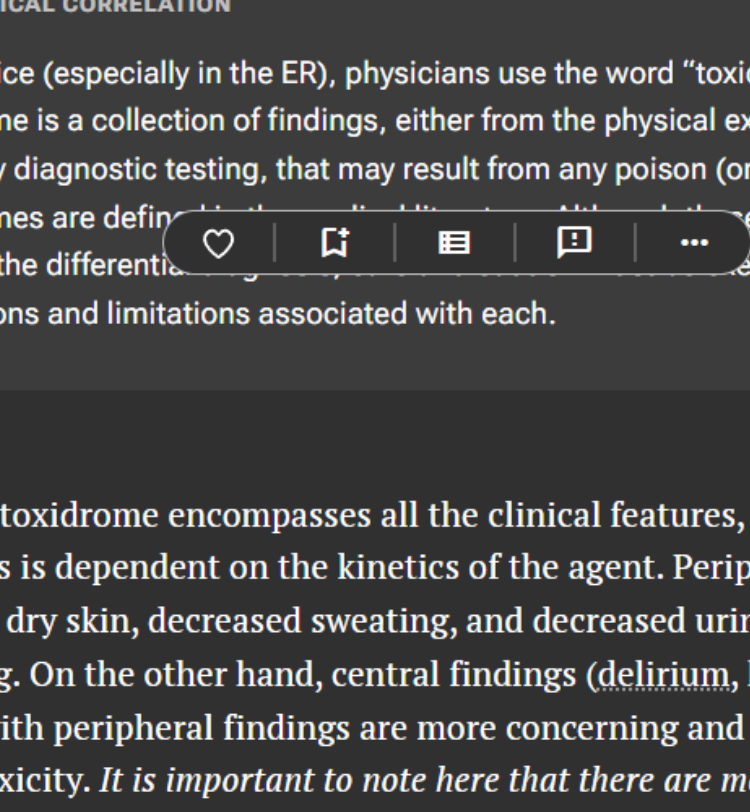


Figure 3: Anticholinergic Toxidrome

CLINICAL CORRELATION

In practice (especially in the ER), physicians use the word “toxidrome.” A toxidrome is a collection of findings, either from the physical examination or from ancillary diagnostic testing, that may result from any poison (or toxin). Numerous toxidromes are defin[illegible]e toxidromes help narrow the differenti[illegible]ercised to realize the exceptions and limitations associated with each.

While the toxidrome encompasses all the clinical features, the presentation of these signs is dependent on the kinetics of the agent. Peripheral findings of mydriasis, dry skin, decreased sweating, and decreased urination alone are less concerning. On the other hand, central findings (delirium, hallucinations) alone or with peripheral findings are more concerning and point toward chronic toxicity. *It is important to note here that there are medications, both prescribed and OTC, belonging to other drug classes but they exhibit anti-cholinergic effects, increasing the risk of adverse effects. We will revisit this concept of increasing anti-cholinergic effect in many other bricks.*

CLINICAL CORRELATION

Use of anti-cholinergics can cause urinary retention and constipation in the elderly. In case of drugs used for urinary incontinence, use in elderly patients (>65 year-old) should be avoided if possible or carefully monitored due to increased risk of constipation.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to NK, what do you tell him about the mechanisms behind urinary retention?

You verify that his allergy medicine contains diphenhydramine, an antihistamine with known anticholinergic properties. These medications can be especially problematic in elderly people, because they have a decrease in cholinergic receptor cells and a slower drug elimination. This puts elderly people at high risk of confusion, urinary retention, constipation, and falls. You tell NK to call for advice before taking any over-the-counter medication and arrange a follow-up visit in 2 days.

Summary

- Muscarinic antagonists (MAs) block the muscarinic receptors, the targets of the parasympathetic nervous system; MAs have many effects, but in general, they decrease secretions, increase heart rate, relax smooth muscles, and slow or adversely affect cognition.
- Important muscarinic antagonists to know include atropine, benztropine, dicyclomine, glycopyrrolate, ipratropium, oxybutynin, scopolamine, tiotropium, trihexyphenidyl, tolterodine, and tropicamide.
- Many MAs, especially atropine, are used in surgery to decrease secretions and reduce the likelihood of bradycardia.
- Inhaled MAs, such as ipratropium and tiotropium, are used to reduce secretions and induce bronchodilation in patients with asthma and COPD.
- MAs are used as eye drops to induce mydriasis and cycloplegia, allowing for better ophthalmoscopic (aka, funduscopic, fundoscopic) eye exams.
- MAs help relax excretory muscles and are thus useful in treating urge incontinence of the bladder and diarrhea, especially in irritable bowel syndrome.

Review Questions

1. Which effect is caused by anticholinergics?

A. ● Decrease in secretions
B. ○ Increase in GI motility
C. ○ Increase in salivation
D. ○ Decrease in Heart rate

Hide Explanation

The correct answer is decreased glandular secretions (A). Anticholinergics inhibit muscarinic receptors, leading to reduced secretions. Increased GI motility (B) and increased salivations (C) are effects of cholinergics, not anticholinergics. Decrease heart rate (E) is a parasympathetic effect or cholinergic effect.

2. Which anticholinergic is commonly used to treat motion sickness?

A. ○ Pilocarpine
B. ● Scopolamine
C. ○ Ipratropium
D. ○ Darifenacin
E. ○ Oxybutynin

Hide Explanation

The correct answer is scopolamine (B). Scopolamine is typically used in the form of a transdermal patch to treat motion sickness and postsurgical nausea. Atropine (A) is used for respiratory secretions, cardiac conduction, and pupil dilation. Ipratropium (C) and Darifenacin (D) are used for asthma and urinary incontinence respectively. Oxybutynin (E) is used for urinary incontinence.

3. From the list below, what could a likely adverse effect of anticholinergics?

A. ○ Diarrhea
B. ○ Salivation
C. ○ Urination
D. ● Constipation

Hide Explanation

The answer is constipation (D). Use of anticholinergics can decrease GI/Intestinal motility leading to co[illegible] Urination (C) are potential adverse effects of cholinergics.

The answer is constipation (D). Use of anticholinergics can decrease GI/Intestinal motility leading to constipation. Diarrhea (A), Salvation (B), and Urination (C) are potential adverse effects of cholinergics.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Predict the organ-specific effects resulting from activation and inhibition of cholinergic receptors based on the tissue expression patterns.~~
02 ☑ ~~Explain the therapeutic uses of cholinergics and anticholinergics based on their target organs and receptor selectivity.~~
03 ☑ ~~Compare and discuss the adverse effects of various anticholinergic agents.~~

Objective Confidence: 3/3